ExamGOAL
PYQ PDF

Free · With answer key

GATE ECE · ENGINEERING MATHEMATICS · CHAPTER-WISE PYQS

Complex Variable

GATE ECE Engineering Mathematics · Previous Year Questions with Answer Key

30	22	2006 – 2025	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	14 questions
Marks 2	16 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Marks 1 14 questions

Q1. GATE ECE 2023

The value of the contour integral, $\oint_C \left(\frac{z+2}{z^2+2z+2}\right)dz$, where the contour C is $\{z : |z+1-\frac{3}{2}j| = 1\}$, taken in the counter clockwise direction, is

(A) $-\pi(1+j)$ (B) $\pi(1+j)$ (C) $\pi(1-j)$ (D) $-\pi(1-j)$

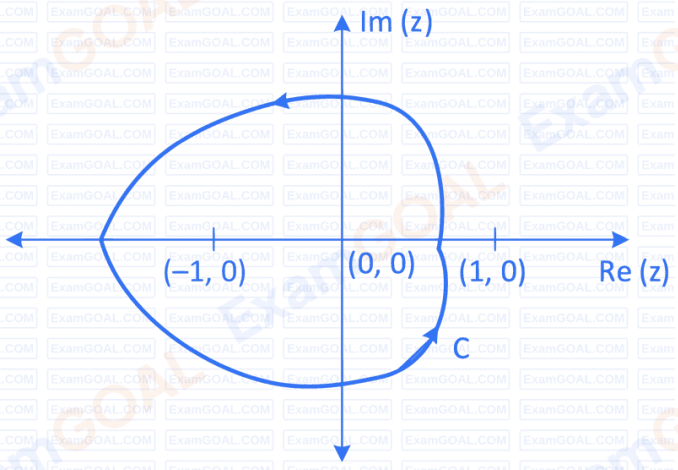
Open solution

Q2. GATE ECE 2022

A simple closed path C in the complex plane is shown in the figure. If

$\oint_c \frac{2^z}{z^2-1}dz = -i\pi A,$

where $i = \sqrt{-1}$, then the value of A is _____ (rounded off to two decimal places).



Open solution

Q3. GATE ECE 2017 Set 2

The residues of a function $f(z) = \frac{1}{(z-4)(z+1)^3}$ are

(A) $\frac{-1}{27}$ and $\frac{-1}{125}$ (B) $\frac{1}{125}$ and $\frac{-1}{125}$
(C) $\frac{-1}{27}$ and $\frac{1}{5}$ (D) $\frac{1}{125}$ and $\frac{-1}{5}$

Open solution

Q4. GATE ECE 2016 Set 3

For $f(z) = \frac{\sin(z)}{z^2}$, the residue of the pole at $z = 0$ _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q5. GATE ECE 2014 Set 1

C is a closed path in the $z-$plane given by $|z| = 3$. The value of the integral $\oint_c \frac{z^2-z+4j}{z+2j} dz$ is

(A) $-4\pi(1+j2)$ (B) $4\pi(3-j2)$ (C) $-4\pi(3+j2)$ (D) $4\pi(1-j2)$

Open solution

Q6. GATE ECE 2014 Set 2

The real part of an analytic function $f(z)$ where $z = x + jy$ is given by $e^{-y}\cos(x)$. The imaginary part of $f(z)$ is

(A) $e^{y}\cos(x)$ (B) $e^{-y}sin(x)$ (C) $-e^{y}sin(x)$ (D) $-e^{-y}sin(x)$

Open solution

Q7. GATE ECE 2011

The value of the integral $\oint_c \frac{-3z+4}{z^2+4z+5} dz$, when C is the circle $|z| = 1$ is given by

(A) 0 (B) $\frac{1}{10}$ (C) $\frac{4}{5}$ (D) 1

Open solution

Q8. GATE ECE 2010

The residues of a complex function $X(z) = \frac{1-2z}{z(z-1)(z-2)}$ at it poles

(A) $\frac{1}{2}, -\frac{1}{2}, 1$ (B) $\frac{1}{2}, -\frac{1}{2}, -1$
(C) $\frac{1}{2}, 1, -\frac{3}{2}$ (D) $\frac{1}{2}, -1, \frac{3}{2}$

Open solution

Q9. GATE ECE 2009

If $f(z) = C_0 + C_1 z^{-1}$ then $\oint_{|z|=1} \frac{1+f(z)}{z} dz$ is given

(A) $2\pi C_1$ (B) $2\pi(1+C_0)$ (C) $2\pi j C_1$ (D) $2\pi j(1+C_0)$

Open solution

Q10. GATE ECE 2008

The residue of the function $f(z) = \frac{1}{(z+2)^2(z-2)^2}$ at z = 2 is

(A) $-\frac{1}{32}$ (B) $-\frac{1}{16}$ (C) $\frac{1}{16}$ (D) $\frac{1}{32}$

Open solution

Q11. GATE ECE 2008

The equation sin(z) = 10 has

(A) no real (or) complex solution (B) exactly two distinct complex solutions
(C) a unique solution (D) an infinite no of complex solutions

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q12. GATE ECE 2007

The value of $\oint_C \frac{1}{(1+z^2)} dz$ where C is the contour $\left|z - \frac{i}{2}\right| = 1$ is

(A) $2\pi i$ (B) π (C) $\tan^{-1}(z)$ (D) $\pi i \tan^{-1}(z)$

Open solution

Q13. GATE ECE 2006

The value of the counter integral

$$\int_{|z-j|=2} \frac{1}{z^2+4} dz \text{ in the positive sense is}$$

(A) $\frac{j\pi}{2}$ (B) $\frac{-\pi}{2}$ (C) $\frac{-j\pi}{2}$ (D) $\frac{\pi}{2}$

Open solution

Q14. GATE ECE 2006

For the function of a complex variable w = lnz (where w = u + jv and z = x + jy) the u = constant lines get mapped in the z-plane as

(A) set of radial straight lines (B) set of concentric circles

(C) set of confocal hyperbola (D) set of confocal ellipses

Open solution

Marks 2 16 questions

Q15. GATE ECE 2025

Which of the following statements involving contour integrals (evaluated counter-clockwise) on the unit circle C in the complex plane is/are TRUE?

(A) $\oint_C e^z dz = 0$ (B) $\oint_C z^n dz = 0$, where n is an even integer

(C) $\oint_C \cos z dz \neq 0$ (D) $\oint_C \sec z dz \neq 0$

Open solution

Q16. GATE ECE 2024

Let z be a complex variable. If $f(z) = \frac{\sin(\pi z)}{z^2(z-2)}$ and C is the circle in the complex plane with $|z| = 3$ then $\oint_C f(z) dz$ is _____.

(A) $\pi^2 j$ (B) $j\pi\left(\frac{1}{2} - \pi\right)$

(C) $j\pi\left(\frac{1}{2} + \pi\right)$ (D) $-\pi^2 j$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q17. GATE ECE 2022

Open solution

Consider the following series :

$$\sum_{n=1}^{\infty} \frac{n^d}{c^n}$$

For which of the following combinations of c, d values does this series converge?

(A) c = 1, d = -1 (B) c = 2, d = 1 (C) c = 0.5, d = -10 (D) c = 1, d = -2

Q18. GATE ECE 2021

Open solution

Consider the integral

$$\oint \frac{\sin(x)}{x^2(x^2+4)} dx$$

where C is counter-clockwise oriented circle defined as |x $-$ i| = 2. The value of the integral is

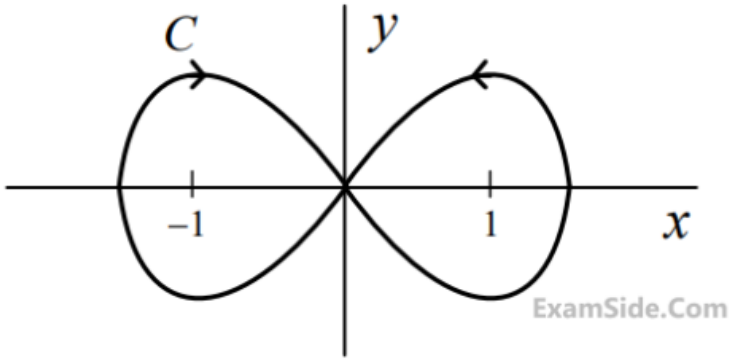
(A) $\frac{\pi}{4}\sin(2i)$ (B) $-\frac{\pi}{8}\sin(2i) + \frac{\pi i}{4}$

(C) $\frac{\pi}{8}\sin(2i)$ (D) $-\frac{\pi}{4}\sin(2i)$

Q19. GATE ECE 2018

Open solution

The contour C given below is on the complex plane $z = x + jy$, where $j = \sqrt{-1}$.



The value of the integral $\frac{1}{\pi j} \oint_C \frac{dz}{z^2-1}$ is _____.

Q20. GATE ECE 2017 Set 2

Open solution

An integral I over a counter clock wise circle C is given by $\mathrm{I} = \oint_c \frac{z^2-1}{z^2+1} e^z \, dz$

If C is defined as $|z| = 3$, then the value of I is

(A) $-\pi i \ \sin(1)$ (B) $-2\pi i \ \sin(1)$

(C) $-3\pi i \ \sin(1)$ (D) $-4\pi i \ \sin(1)$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q21. GATE ECE 2016 Set 2

Consider the complex valued function $f(z) = 2z^3 + b|z|^3$ where z is a complex variable. The value of b for which the function $f(z)$ is analytic is _____.

Open solution

Q22. GATE ECE 2016 Set 1

In the following integral, the contour C encloses the points $2\pi j$ and $-2\pi j$. The value of the integral $-\frac{1}{2\pi}\oint_c \frac{\sin z}{(z-2\pi j)^3}dz$ is _____.

Open solution

Q23. GATE ECE 2016 Set 3

The value of the integral $\frac{1}{2\pi j}\oint_C \frac{e^z}{z-2}dz$ along a closed contour c in anti-clockwise direction for

(i) the point $z_0 = 2$ inside the contour c, and
(ii) the point $z_0 = 2$ outside the contour c, respectively, are

(A) (i) 2.72, (ii) 0 (B) (i) 7.39, (ii) 0 (C) (i) 0, (ii) 2.72 (D) (i) 0, (ii) 7.39

Open solution

Q24. GATE ECE 2015 Set 3

If C is a circle of radius r with centre z_0 in the complex z-plane and if $'n'$ is a non-zero integer, then $\oint_C \frac{dz}{(z-z_0)^{n+1}}$ equals

(A) $2\pi nj$ (B) 0 (C) $\frac{nj}{2\pi}$ (D) $2\pi n$

Open solution

Q25. GATE ECE 2015 Set 2

Let $f(z) = \frac{az+b}{cz+d}$. If $f(z_1) = f(z_2)$ for all $z_1 \neq z_2$. $a = 2$, $b = 4$ and $C = 5$, then d should be equal to

Open solution

Q26. GATE ECE 2015 Set 2

If C denotes the counter clockwise unit circle. The value of the contour integral $\frac{1}{2\pi i}\oint_c \mathrm{Re}\{z\}dz$ is _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q27. GATE ECE 2015 Set 1

Let $z = x + iy$ be a complex variable. Consider that contour integration is performed along the unit circle in anticlockwise direction . Which one of the following statement is NOT TRUE?

Open solution

(A) The residue of $\frac{z}{z^2-1}$ at $z = 1$ is $\frac{1}{2}$

(B) $\oint_C z^2\ dz = 0$

(C) $\frac{1}{2\pi i} \oint_c \frac{1}{z} dz = 1$

(D) $\overline{z}$ (complex conjugate of z) is an analytical function.

Q28. GATE ECE 2012

If $x = \sqrt{-1},$ then the value of X^x is

Open solution

(A) $e^{-\pi/2}$ (B) $e^{\pi/2}$ (C) x (D) 1

Q29. GATE ECE 2012

Given $f(z) = \frac{1}{z+1} - \frac{2}{z+3}$. If C is a counterclockwise path in the z-plane such that $|z+1| = 1$, the value of $\frac{1}{2\,\pi\,j} \oint_c f(z)dz$ is

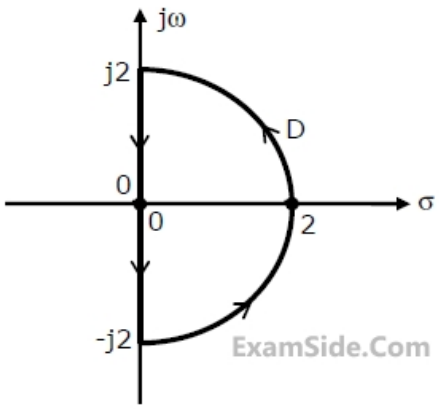
Open solution

(A) -2 (B) -1 (C) 1 (D) 2

Q30. GATE ECE 2007

If the semi-circular contour D of radius 2 is as shown in the figure, then the value of the integral $\oint_D \frac{1}{s^2-1} ds$ is

Open solution



(A) $i\pi$ (B) $-i\pi$ (C) $-\pi$ (D) π

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

Marks 1

1. **B**	2. **0.5**	3. **B**	4. **1**	5. **C**
6. **B**	7. **A**	8. **C**	9. **D**	10. **A**
11. **D**	12. **B**	13. **D**	14. **B**	

Marks 2

15. **A, B**	16. **D**	17. **B, D**	18. **B**	19. **2 to 2**
20. **D**	21. **0**	22. **- 133.8**	23. **B**	24. **B**
25. **10**	26. **0.5**	27. **D**	28. **A**	29. **C**
30. **A**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs